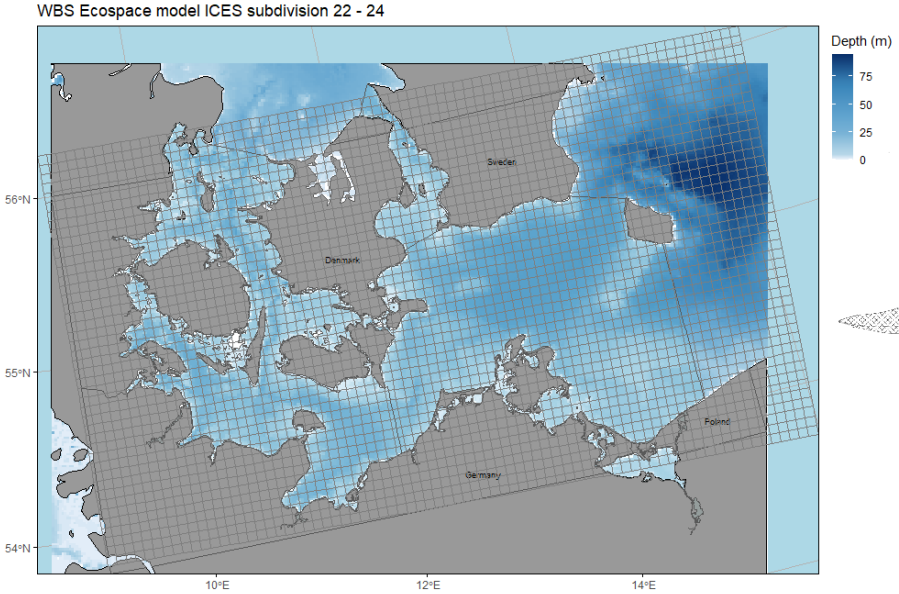
HELMHOLTZ

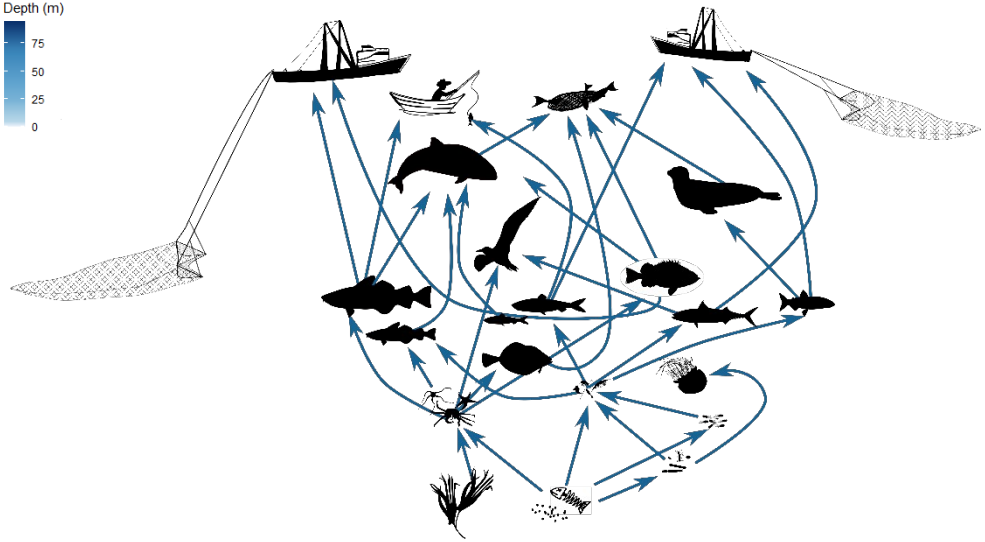
GEOMAR

Fisheries and climate shape the future of western Baltic species – a spatially explicit trophic approach

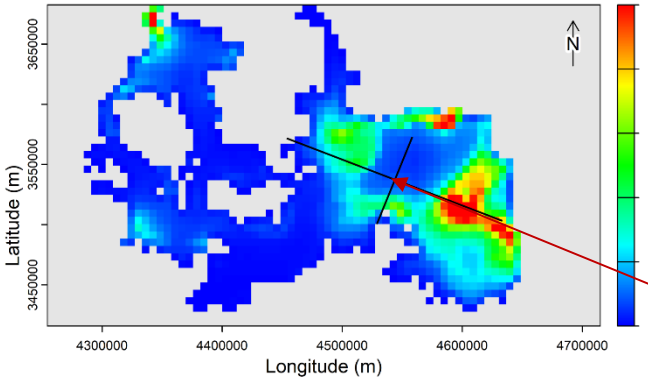
Marcela C. Nascimento and Marco Scotti

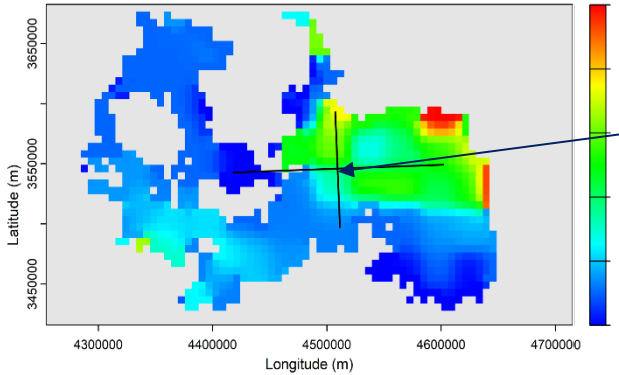
The western Baltic Sea Ecospace model





Model validation – cod distribution

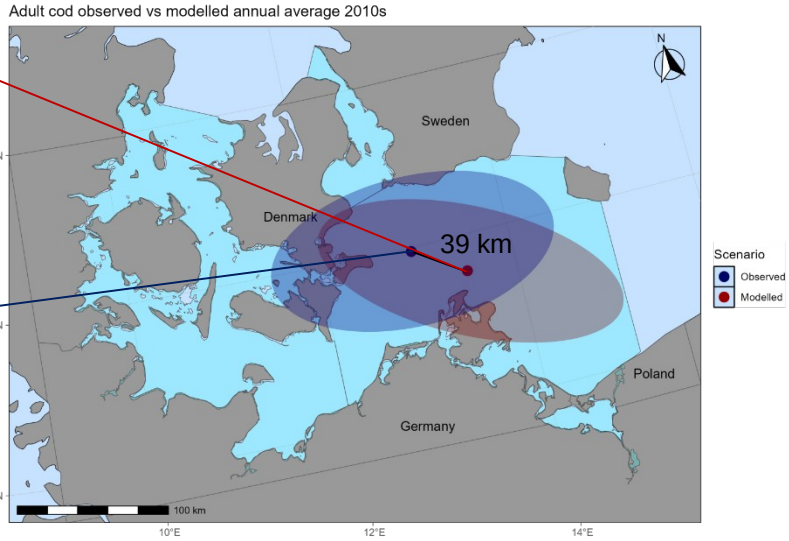




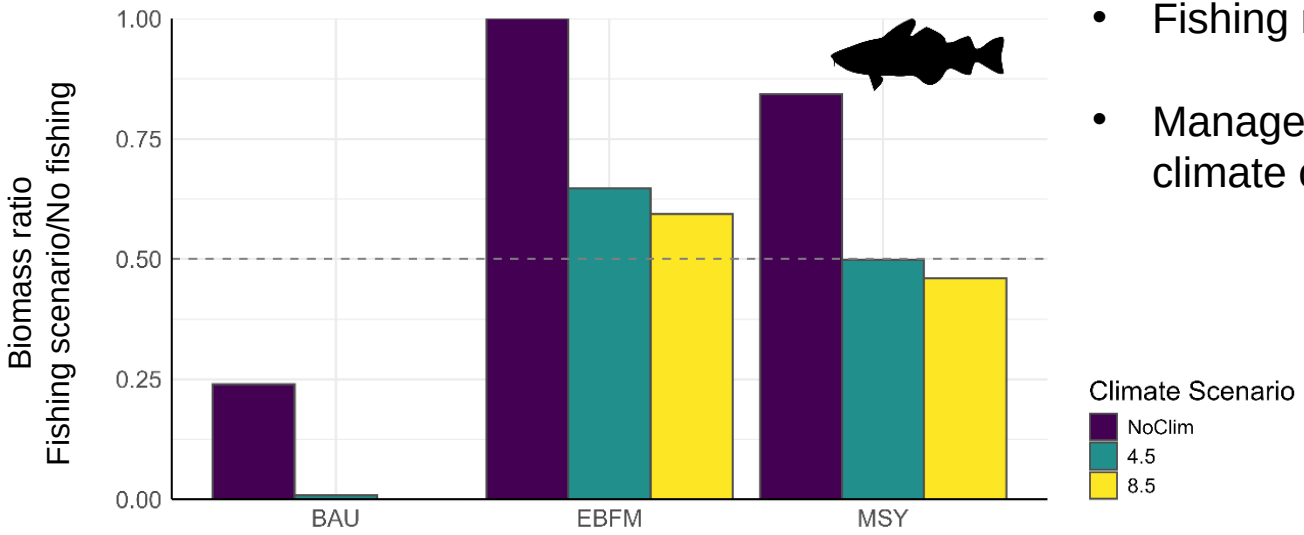


Adult cod

2010's



Biomass predictions – 2065

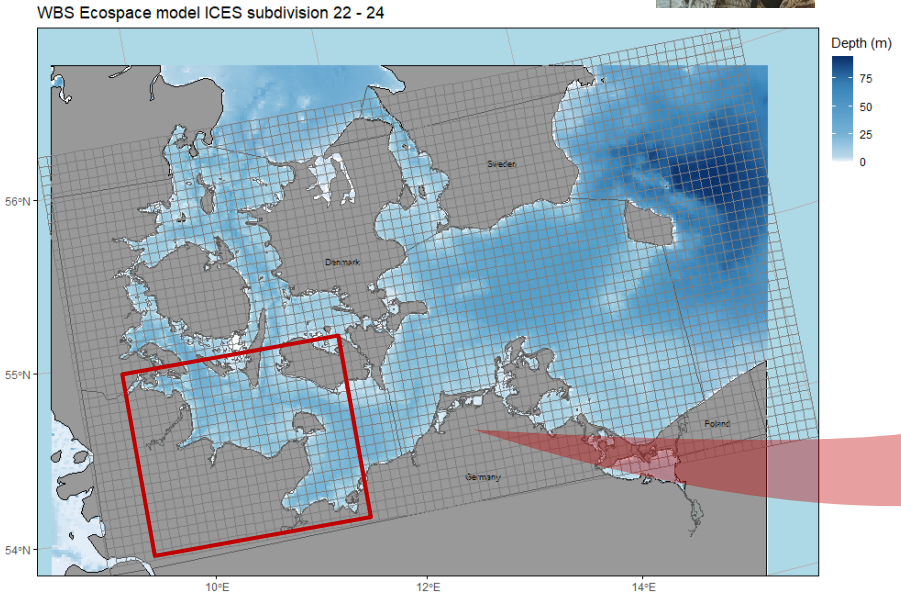


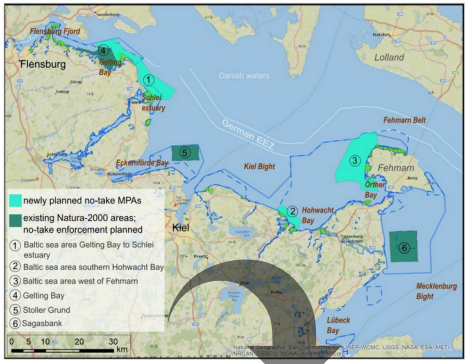
- Fishing main driver of change
- Management can buffer climate change effects

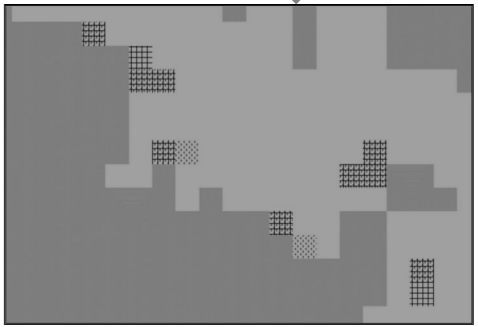
MPAs in Schleswig-Holstein

Smilla Todt Master's thesis

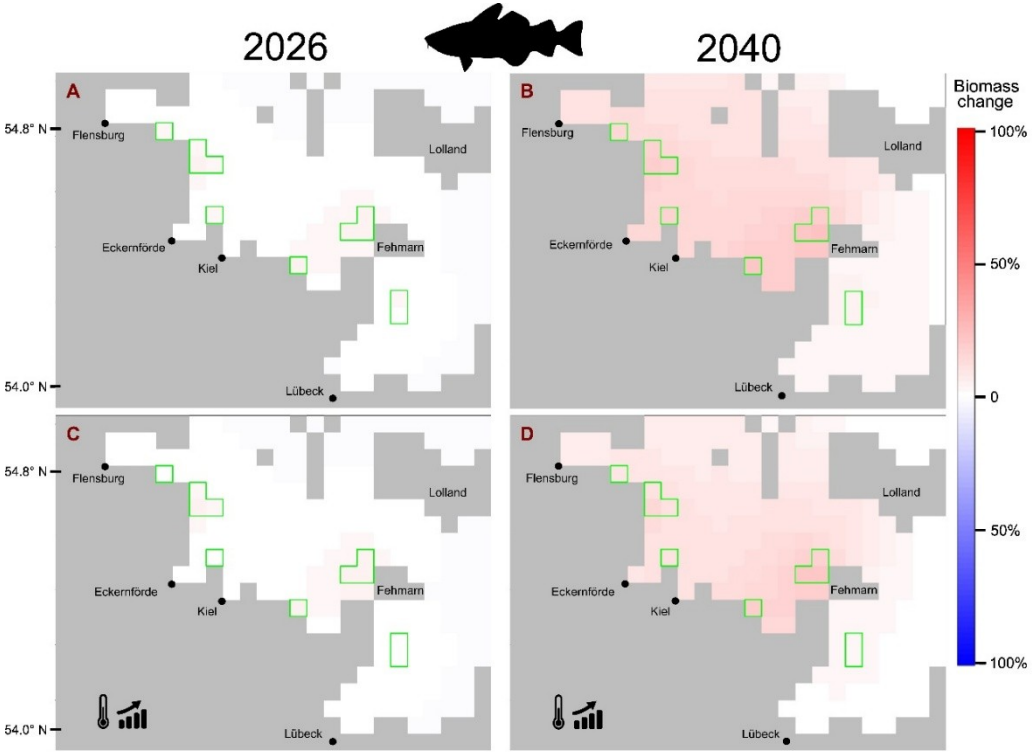








MPAs in Schleswig-Holstein – biomass

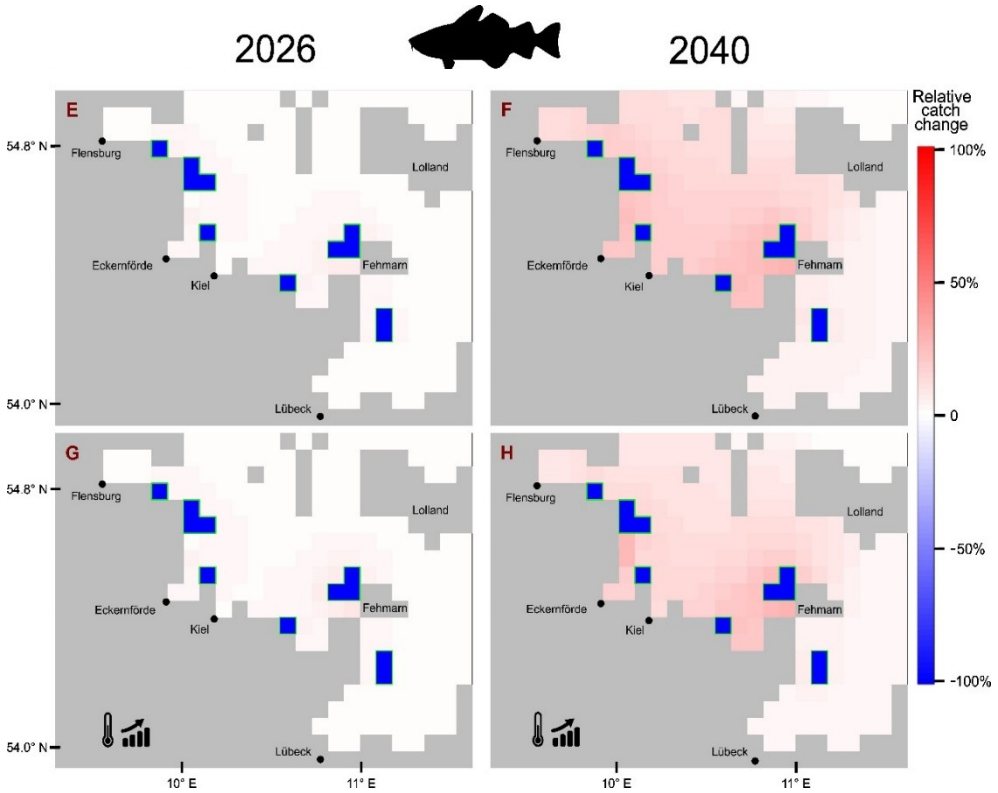


No climate change

RCP 8.5



MPAs in Schleswig-Holstein – catch



No climate change

RCP 8.5



Fishery & climate change – conclusions

- Fishery is the main force shaping the WBS stocks
- WBS commercial stocks are negatively impacted by climate change
- Sustainable fisheries management (i.e., EBFM) buffers partially the negative impacts of climate change
- MPAs positively influence biodiversity and fishery yield